OntoCoDec: Near-Complete Type-Valid Information Extraction via Compiled Ontology Constraints

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Abstract

LLM-based information extraction (IE) systems produce ontology constraint violations at non-trivial rates: 15.4% of triples on SciERC and 18.7% on DocRED violate domain/range type constraints under zero-shot prompting. Schema-informed prompting reduces violations by 62% on a small ontology (SciERC, 7 relations) but only 32% on a complex one (DocRED, 96 relations), and individual relation types can regress. We present OntoCoDec, a framework that automatically compiles IE ontology schemas into two composable constrained-decoding layers: (1) a grammar layer enforcing JSON structure and type vocabulary via context-free grammars, and (2) a semantic layer enforcing cross-field domain/range constraints through a stateful entity registry with dynamic logit masking. On SciERC, OntoCoDec reduces domain/range violations from 15.4% to 0.93% (↓94%) with neutral F1 impact (+0.13pp, p=0.58). On DocRED, violations decrease from 18.5% to 10.2% (↓45%). Cross-model validation on Llama-3.1-8B confirms generalization without code modification (SciERC DR↓99.4%, DocRED DR↓90.2%). The method introduces cross-field consistency (CF) violations (up to 9.9% on SciERC, 12.0% on Llama DocRED) as an inherent trade-off of per-relation type enforcement, which we analyze as a fundamental constraint of forward-only decoding.

1 Introduction

Large language models (LLMs) increasingly serve as zero-shot information extraction systems, converting unstructured text into structured knowledge representations without task-specific training. However, LLM outputs frequently violate the ontological constraints that define valid extractions. When a schema specifies that the `Feature-of` relation requires a METHOD head and a TASK tail, an LLM may instead produce a MATERIAL–TASK pair—a domain/range (DR) violation that renders the triple structurally invalid for downstream knowledge graph construction.

We present the first systematic measurement of ontology constraint violations in LLM-based IE across two benchmarks spanning an order of magnitude in ontology complexity. Our violation landscape study reveals that DR violation rates are driven by *constraint tightness*—the fraction of entity-type pairs permitted by a relation's domain/range specification—rather than ontology size. SciERC (7 relations) and DocRED (96 relations) exhibit comparable per-triple DR rates (15.4% vs. 18.7%) despite a 14× difference in relation count.

Prompt-level intervention (explicitly describing constraints in the prompt) reduces but cannot eliminate violations: overall DR drops 62% on SciERC but only 32% on DocRED, and one relation type (`Evaluate-for`) *increases* in violation rate. This motivates decoding-time constraint enforcement.

OntoCoDec automatically compiles an IE ontology schema into two composable constraint layers:

1. A **grammar layer** (built on context-free grammar constrained decoding) that enforces JSON structure, relation vocabulary, and entity-type vocabulary.

2. A **semantic layer** that maintains a stateful entity registry during generation and dynamically masks logits to enforce domain/range constraints.

The layers compose via bitwise AND of token masks with an enabling dependency: the grammar layer ensures every generated relation name is in the valid vocabulary, which enables the semantic layer to

look up and enforce the corresponding domain/range rules. When the intersection of masks is empty, the system falls back to grammar-only constraints, maintaining structural validity as a guaranteed floor.

Experiments on SciERC and DocRED with Qwen3-8B and Llama-3.1-8B demonstrate that OntoCoDec reduces DR violations by 45–99% depending on the dataset–model combination, eliminates relation-vocabulary and entity-type violations entirely, and preserves or improves extraction F1. We identify cross-field consistency (CF) violations—where the same entity receives different type assignments across relations—as an inherent trade-off of per-relation enforcement, occurring at rates of 2.6–12.0%.

2 Background

IE Ontology Structure. An IE ontology $\mathcal{O} = (\mathcal{E}, \mathcal{R}, \mathcal{C})$ consists of entity types $\mathcal{E}$, relation types $\mathcal{R}$, and domain/range constraints $\mathcal{C} : \mathcal{R} \to 2^{\mathcal{E}} \times 2^{\mathcal{E}}$. For each relation r, $\mathcal{C}(r) = (\text{dom}(r), \text{rng}(r))$ specifies the set of valid head entity types and tail entity types, respectively.

Violation Types. We define four violation categories for a predicted triple (h, t_h, r, t, t_t) where h, t are entity mentions, t_h, t_t are entity types, and r is a relation:

- **Domain/Range (DR)**: $t_h \notin \text{dom}(r)$ or $t_t \notin \text{rng}(r)$.
- **Relation Vocabulary (RV)**: $r \notin \mathcal{R}$.
- **Entity Type (ET)**: $t_h \notin \mathcal{E}$ or $t_t \notin \mathcal{E}$.
- **Cross-Field Consistency (CF)**: The same entity mention is assigned different types in different triples within the same document.

Constraint Tightness. We define constraint tightness $\tau(r) = 1 - \frac{|\text{dom}(r)| \cdot |\text{rng}(r)|}{|\mathcal{E}|^2}$ as the fraction of entity-type pairs that are *invalid* for relation r. Higher tightness means fewer valid type combinations, predicting higher violation rates.

Grammar-Constrained Decoding. Grammar-constrained decoding restricts autoregressive generation to strings accepted by a formal grammar (typically a CFG) by masking logits at each decoding step. Recent systems such as XGrammar achieve this efficiently through precomputed token masks. This approach guarantees structural validity (e.g., valid JSON) but cannot express cross-field semantic constraints.

3 Violation Landscape Study

We measure ontology constraint violations produced by Qwen3-8B under zero-shot prompting on SciERC (7 relations, 6 entity types) and DocRED (96 relations, 7 entity types).

3.1 Baseline Violation Rates

Table 1 reports per-triple violation rates under minimal prompting (task description only) and schema-informed prompting (explicit constraint descriptions appended).

Key Findings. (1) DR violation rates are comparable across ontologies of vastly different sizes (15.4% vs. 18.7%), consistent with the tightness hypothesis. (2) Schema-informed prompting achieves 62% DR reduction on SciERC but only 32% on DocRED, indicating that prompt-level intervention scales poorly with ontology complexity. (3) DocRED exhibits 28.9% RV violations—nearly one-third of predicted relations use non-existent relation names—reflecting the difficulty of zero-shot generation over large relation vocabularies.

Table 1: Ontology violation rates (%) under zero-shot prompting. DR = domain/range, RV = relation vocabulary, ET = entity type. Δ = relative change from minimal to schema-informed.

Dataset	Prompt	DR	RV	ET	Triples
SciERC	Minimal	15.4	1.4	0.4	1724
	Schema	5.8	—	—	1661
	Δ	↓62.4%	—	—	—
DocRED	Minimal	18.7	28.9	7.5	16540
	Schema	12.8	18.4	—	—
	Δ	↓32.0%	↓36.3%	—	—

3.2 Per-Relation Analysis

On SciERC, violations concentrate on relations with tight constraints. Under schema-informed prompting, `Feature-of` DR drops by 22.3 percentage points (pp) and `Compare` by 14.8pp, but `Evaluate-for` *increases* by 5.8pp. This regression illustrates a prior-constraint conflict: the model's distributional prior (e.g., METHOD→TASK for `Evaluate-for`) conflicts with the explicit constraint signal, destabilizing type assignments on boundary cases.

On DocRED, the relation `P31_instance_of` exhibits 60.4% DR with 533 triples, disproportionately driving the aggregate rate. Relation-F1 under minimal prompting is 8.23% (SciERC) and 7.74% (DocRED), reflecting the inherent difficulty of zero-shot IE rather than the quality of the constraint analysis.

4 The OntoCoDec Framework

OntoCoDec takes an IE ontology definition $\mathcal{O}$ and automatically compiles it into decoder constraints that enforce type validity during autoregressive generation.

4.1 Ontology Compiler

The compiler produces two artifacts from $\mathcal{O}$:

1. A **context-free grammar** G encoding JSON structure, the relation vocabulary $\mathcal{R}$, and entity-type vocabulary $\mathcal{E}$. This grammar is compiled into precomputed token masks via XGrammar.
2. A **constraint table** $T : \mathcal{R} \to 2^{\mathcal{E}} \times 2^{\mathcal{E}}$ mapping each relation to its valid head/tail entity types, used by the semantic layer at runtime.

4.2 Dual-Layer Constraint Architecture

Grammar Layer. At each decoding step, the grammar layer computes a token mask M_{gram} restricting generation to tokens consistent with G. This eliminates RV and ET violations by construction.

Semantic Layer (TypeConstraintProcessor). The semantic layer maintains a stateful entity registry $\mathcal{S}$ that tracks entity types assigned during generation. When the decoder reaches a type-position token (e.g., the head entity type field of a triple), the semantic layer:

1. Identifies the current relation r from the partially generated output.
2. Computes M_{sem} by allowing only entity types in $\text{dom}(r)$ (for head) or $\text{rng}(r)$ (for tail).
3. Updates $\mathcal{S}$ with the generated type assignment.

Table 2: SciERC ablation (Qwen3-8B). DR/RV/ET/CF are per-triple violation rates (%). R-F1 and E-F1 are relation and entity micro-F1 (%).

Config	DR	RV	ET	CF	R-F1	E-F1	Triples
Unconstrained	15.4	1.4	0.4	0.1	8.23	31.01	1724
Grammar-only	14.8	0.0	0.0	0.1	8.23	30.98	—
Semantic-only	11.7	1.5	0.4	2.7	8.15	31.01	—
Full S1	0.9	0.0	0.0	9.9	8.36	30.98	1729
Full S2	1.0	0.0	0.0	8.0	8.03	31.90	1644

Layer Composition. The final token mask is $M = M_{\text{gram}} \wedge M_{\text{sem}}$. When $M = \emptyset$ (the semantic and grammar constraints conflict), the system falls back to M_{gram}, maintaining structural validity. This fallback is essential for graceful degradation on complex ontologies—on Llama DocRED, fallback occurs approximately 57 times per document.

Decidability Taxonomy. Constraints are classified by prefix-decidability: (a) *Purely local*: entity type $\in \mathcal{E}$ (grammar layer); (b) *Prefix-decidable with state*: domain/range compatibility given partial output (semantic layer, requires entity registry); (c) *Truly non-local*: forward coreference consistency—out of scope for forward-only decoding.

4.3 Decoding Strategies

S1 (Single-Pass). Both layers are applied throughout generation. This is the primary strategy.

S2 (Two-Pass). Pass 1 extracts entities with grammar-only constraints. Pass 2 generates relation triples with an input-dependent grammar compiled from Pass 1 entities. S2 is an architectural variant demonstrating framework flexibility; it does not substantially differ from S1 on any metric (Section 5.5).

5 Experiments

5.1 Setup

We evaluate on SciERC (500 abstracts, 7 relations, 6 entity types) and DocRED (998 documents, 96 relations, 7 entity types) under zero-shot extraction. Primary model: Qwen3-8B. Cross-model validation: Llama-3.1-8B. F1 is reported as a control variable verifying that constraint enforcement does not degrade extraction quality, not as a claim of state-of-the-art performance.

5.2 SciERC Ablation

Table 2 presents the full ablation across five configurations on SciERC.

Grammar Layer. Grammar-only eliminates RV (1.4%→0%) and ET (0.4%→0%) but has negligible effect on DR (15.4%→14.8%), as expected—CFGs cannot express cross-field constraints. R-F1 is unchanged (8.23%).

Semantic Layer. Semantic-only reduces DR by 24% (15.4%→11.7%) but introduces CF violations (2.7%). Without the grammar layer, the semantic layer encounters out-of-vocabulary relation names and cannot look up constraint rules, explaining its limited DR reduction.

Table 3: DocRED ablation (Qwen3-8B). Parse failures excluded from F1 computation (12–13 documents).

Config	DR	RV	ET	CF	R-F1	E-F1	Triples
Unconstrained	18.5	29.7	2.5	0.1	7.74	58.15	17497
Grammar-only	26.2	0.03	0.0	0.1	8.31	59.38	—
Semantic-only	13.8	29.9	4.2	1.1	7.70	57.81	—
Full S1	10.2	0.03	0.0	2.6	8.23	59.37	—

Table 4: Cross-model validation (Llama-3.1-8B). OntoCoDec applied without code modification. Parse failure rates in parentheses where non-trivial.

Dataset	Config	DR	RV	ET	CF	R-F1	E-F1
SciERC	Unconstr.	13.3	0.6	0.0	0.5	2.68	23.56
	Full S1	0.1	0.0	0.0	6.6	2.54	23.49
DocRED	Unconstr. (6.8% parse fail)	26.1	16.0	0.1	0.2	5.98	55.65
	Full S1 (9.1% parse fail)	2.6	0.0	0.0	12.0	6.27	55.01

Full System (S1). With both layers, DR drops 94% (15.4%→0.9%). RV and ET are eliminated entirely. R-F1 is slightly positive (+0.13pp) but not statistically significant (p=0.58, bootstrap CI with 10,000 iterations). CF rises to 9.9%—an inherent trade-off of per-relation independent type enforcement (Section 6.1).

5.3 DocRED Ablation

Table 3 presents the DocRED ablation.

Grammar Layer Increases DR. Grammar-only *increases* DR from 18.5% to 26.2% (+7.7pp). When grammar forces out-of-vocabulary relation names (29.7% of baseline triples) into valid relation names, the forced valid name carries different domain/range constraints than the model intended, creating new DR violations. This effect is specific to ontologies with high baseline RV—grammar exposes latent DR violations that were previously masked by invalid relation names.

Full System. Full S1 reduces DR to 10.2% (↓45%), substantially less than SciERC's 94% reduction. The gap is explained by fallback frequency: on complex ontologies with tight constraints, semantic mask∩grammar mask more frequently yields $\emptyset$, triggering grammar-only fallback. Grammar-only DR is 26.2%, so frequent fallback leaves residual DR elevated. R-F1 improves to 8.23% (+0.50pp, $p<0.001$), driven by grammar elimination of OOV relation names.

5.4 Cross-Model Validation

Table 4 presents cross-model results with Llama-3.1-8B, applied with zero code modification.

SciERC. DR drops from 13.3% to 0.1% (↓99.4%). Llama's low baseline RV (0.6% vs. Qwen's 1.4%) means grammar encounters fewer OOV issues, enabling cleaner semantic enforcement. R-F1 decreases slightly (−0.14pp) on an already low baseline of 2.68%. CF reaches 6.6%, lower than Qwen's 9.9% on the same dataset.

Table 5: Post-hoc filtering vs. OntoCoDec (Qwen3-8B). Rej. = fraction of triples rejected by post-hoc filter. TP = true positive triples.

Dataset	Method	R-F1	Recall	TP	Rej.	ET res.
SciERC	Post-hoc	8.63	10.68	104	16.8%	—
	OntoCoDec	8.36	11.60	—	0%	0%
DocRED	Post-hoc	9.61	8.40	1010	48.1%	2.5%
	OntoCoDec	8.23	10.02	—	0%	0%

DocRED. DR drops from 26.1% to 2.6% (↓90.2%)—substantially better than Qwen's 45% reduction on the same dataset. This cross-model asymmetry is explained by the interaction between baseline RV and DR reduction: Qwen's higher RV baseline (29.7% vs. 16.0%) means grammar forces more OOV triples into valid names, exposing more latent DR violations that the semantic layer must then resolve.

However, CF reaches 12.0%—the highest of any configuration, and 4.7× Qwen's rate on DocRED (2.6%). Parse failures also increase from 68 to 91 (+33.8%), indicating constrained decoding destabilizes Llama's JSON generation on the complex DocRED ontology. E-F1 drops 0.64pp. The cross-model variability in CF rates (2.6%–12.0%) indicates that CF behavior is model-dependent and difficult to predict.

5.5 S2 Two-Pass Ablation

S2 (two-pass decoding) was designed to mitigate CF by separating entity extraction from relation generation. On full SciERC evaluation (551 documents), S2 achieves DR = 1.0%, CF = 8.0%, R-F1 = 8.03%. Compared to S1 (DR = 0.9%, CF = 9.9%, R-F1 = 8.36%), S2 provides a non-significant 1.9pp CF reduction while generating 85 fewer triples (1644 vs. 1729) and achieving lower R-F1 (−0.33pp). S2 is an architectural variant demonstrating framework flexibility; it does not solve the CF trade-off.

5.6 Post-Hoc Filtering Baseline

We compare OntoCoDec against a post-hoc filtering baseline that removes triples violating domain/range constraints after unconstrained generation.

Post-hoc filtering achieves higher F1 than OntoCoDec on both datasets (SciERC: 8.63 vs. 8.36; DocRED: 9.61 vs. 8.23). In the low-F1 regime, removing incorrect triples (most of which are false positives) substantially boosts precision.

However, post-hoc filtering has three limitations: (1) **Recall loss**: post-hoc recall is lower (SciERC: 10.68 vs. 11.60; DocRED: 8.40 vs. 10.02) because correctly extracted triples with type violations are also removed—on DocRED, 127 true positive triples (11.2% of all TPs) are discarded. (2) **Information coverage**: DocRED post-hoc discards 48.1% of all triples, losing nearly half the extracted information. (3) **Residual ET**: post-hoc only removes triples, not invalid entity types from the entity list—DocRED retains 2.5% ET violations after filtering, while OntoCoDec achieves 0%.

The choice between approaches depends on the application: post-hoc filtering favors precision when false positives are costly; OntoCoDec favors recall and completeness when information coverage matters.

5.7 Statistical Significance

Table 6 reports bootstrap confidence intervals (10,000 iterations) for the F1 difference between unconstrained and OntoCoDec Full S1.

SciERC F1 changes are not statistically significant—OntoCoDec is F1-neutral on SciERC. DocRED shows statistically significant improvements in both Rel F1 (+0.50pp) and Ent F1 (+1.23pp), driven by the grammar layer's elimination of OOV relation names (29.7%→0.03%): under the evaluation method that treats OOV predictions as false positives, removing OOV relations mechanically improves precision.

Table 6: Bootstrap significance tests (10,000 iterations). Δ in percentage points.

Dataset	Metric	Δ (pp)	p-value	95% CI
SciERC	Rel F1	+0.13	0.58	—
	Ent F1	−0.03	0.53	—
DocRED	Rel F1	+0.50	<0.001	[+0.23, +0.81]
	Ent F1	+1.23	<0.001	[+0.94, +1.55]

Table 7: CF violation rates (%) across configurations. CF is absent from unconstrained/grammar-only baselines ($\leq$0.1%) and rises with semantic enforcement.

Config	SciERC		DocRED	
	Qwen	Llama	Qwen	Llama
Unconstrained	0.1	0.5	0.1	0.2
Semantic-only	2.7	—	1.1	—
Full S1	9.9	6.6	2.6	12.0
Full S2	8.0	—	—	—

Grammar-only achieves the same E-F1 as the full system (59.38% vs. 59.37%), confirming that the semantic layer does not contribute to F1.

6 Analysis

6.1 Cross-Field Consistency Violations

CF violations—where the same entity receives different type assignments across relations—are the primary trade-off of OntoCoDec. Table 7 summarizes CF rates across all configurations.

CF is 100% attributable to the semantic layer: grammar-only CF equals unconstrained CF (0.1%), while semantic-only already shows elevated CF (2.7% on SciERC). The mechanism is per-relation independent enforcement: the semantic layer forces entity types to satisfy the *current* relation's constraints without considering types assigned by previous relations.

CF Subtypes. Entity-vs-relation type drift (EVR)—where the same entity has different types in the entity list vs. relation triples—accounts for >90% of CF violations. The dominant conflict patterns (GENERIC↔OTHERSCIENTIFICTERM, MATERIAL↔METHOD) reflect genuine semantic ambiguity in the SciERC type system, where the same term can legitimately belong to multiple types depending on context.

Severity Distinction. DR violations produce structurally invalid triples that violate the ontology definition. CF violations produce individually valid triples whose cross-triple entity typing is inconsistent—a softer form of schema non-compliance. We report both transparently and note that the net violation picture on SciERC is: baseline 15.4% DR → OntoCoDec 0.9% DR + 9.9% CF, a shift from hard violations to soft inconsistencies.

6.2 Constraint Tightness and DR Reduction

The DR reduction ceiling is determined jointly by constraint tightness and the model's baseline RV rate. On DocRED ($\bar{\tau}$=0.842), Qwen achieves DR↓45% while Llama achieves DR↓90%—on the same ontology. The explanation: Qwen's higher RV baseline (29.7% vs. 16.0%) means grammar forces more OOV triples into valid names, exposing latent DR violations. Grammar-only DR *increases* on Qwen

Table 8: DR reduction summary across all model–dataset combinations.

Model	Dataset	DR Base (%)	DR S1 (%)	↓%
Qwen3-8B	SciERC	15.4	0.9	94
Qwen3-8B	DocRED	18.5	10.2	45
Llama-3.1-8B	SciERC	13.3	0.1	99
Llama-3.1-8B	DocRED	26.1	2.6	90

DocRED (18.5%→26.2%) but this effect is less pronounced on Llama, allowing the semantic layer to operate on a cleaner input distribution.

This dual-factor model (tightness × RV baseline) provides a more nuanced understanding than tightness alone: constraint tightness predicts the *difficulty* of enforcement, while RV baseline predicts the *noise floor* after grammar normalization.

6.3 Comparison of DR Reduction Across Settings

DR reduction ranges from 45% (Qwen DocRED) to 99% (Llama SciERC). The method generalizes across model families without code modification, though effectiveness varies substantially with the model–ontology interaction. The Qwen DocRED result (45% reduction, 10.2% residual) is the weakest configuration, leaving approximately one in ten triples with invalid domain/range types.

7 Related Work

Grammar-Constrained Decoding. Structured output generation via grammar-constrained decoding has been explored for code generation and JSON output. XGrammar and related systems enforce CFG-expressible constraints efficiently. OntoCoDec extends this paradigm to cross-field semantic constraints that CFGs cannot express.

Constrained Decoding for NLP. Prior work on constrained decoding for NLP focuses on lexical constraints (e.g., keyword inclusion), length constraints, or structural constraints (e.g., valid parse trees). OntoCoDec addresses a different constraint class: type-level semantic constraints derived from ontology definitions.

Schema-Guided IE. Schema-guided generation and extraction systems use schema definitions to condition the generation process. Unlike prompt-based approaches, OntoCoDec enforces constraints at the logit level, providing deterministic compliance for grammar-expressible constraints and best-effort compliance for semantic constraints.

Post-Hoc Validation. Post-hoc approaches validate and filter extraction outputs against ontology constraints. As shown in Section 5.6, post-hoc filtering trades recall for precision and cannot address entity-type violations within the entity list itself.

8 Discussion

Zero-Shot Evaluation Setting. We deliberately evaluate in a zero-shot setting to demonstrate that OntoCoDec works without task-specific training. Absolute F1 is low (8–9% Rel F1) as expected—supervised SOTA on SciERC relation F1 is 40–50%. Our contribution is the constraint enforcement method, not extraction quality. F1 is reported as a control variable verifying that constraints do not degrade quality.

Limitations. (1) CF violations (2.6–12.0%) are an inherent limitation of per-relation independent enforcement. The two-pass S2 strategy, designed to mitigate CF, achieves only 1.9pp reduction (p not significant). A global entity-type consensus pass is needed as future work. (2) Effectiveness varies substantially across model–ontology combinations (DR↓45–99%), and CF rates are model-dependent (Qwen 2.6% vs. Llama 12.0% on DocRED). (3) Evaluation is limited to two datasets, two 8B-parameter models, and zero-shot extraction. Generalization to supervised settings, larger models, or more diverse ontologies remains to be validated. (4) On DocRED, constrained decoding increases Llama's parse failure rate from 6.8% to 9.1%, indicating stability concerns with certain model–ontology combinations.

9 Conclusion

We present the first systematic measurement of ontology constraint violations in LLM-based IE and introduce OntoCoDec, a framework that compiles ontology schemas into composable grammar and semantic decoding constraints. OntoCoDec reduces domain/range violations by 45–99% across two datasets and two model families, eliminates relation-vocabulary and entity-type violations entirely, and preserves extraction F1. The method introduces cross-field consistency violations (2.6–12.0%) as an inherent trade-off of per-relation enforcement—an open problem for future work on global constraint propagation during autoregressive generation.

References

[placeholder] References omitted for draft review.